

Developing & Maintaining the World's Largest* MiniZinc Model

Presented by Danyal Mirza

Content



- Quick background on Ericsson and RAN networks
- Using combinatorial optimization in product configuration
- Developing, executing & maintaining very large MiniZinc models

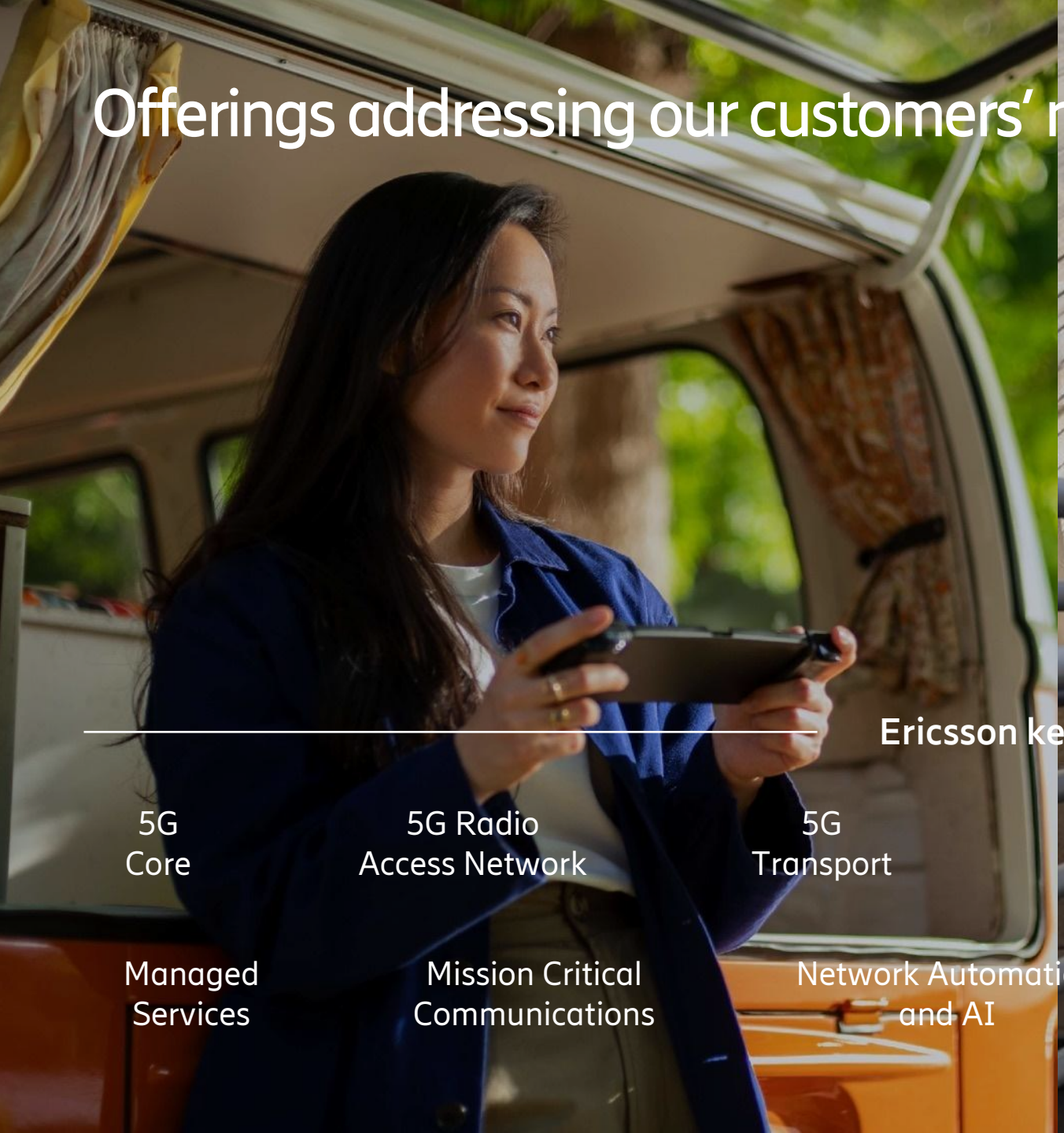
Danyal

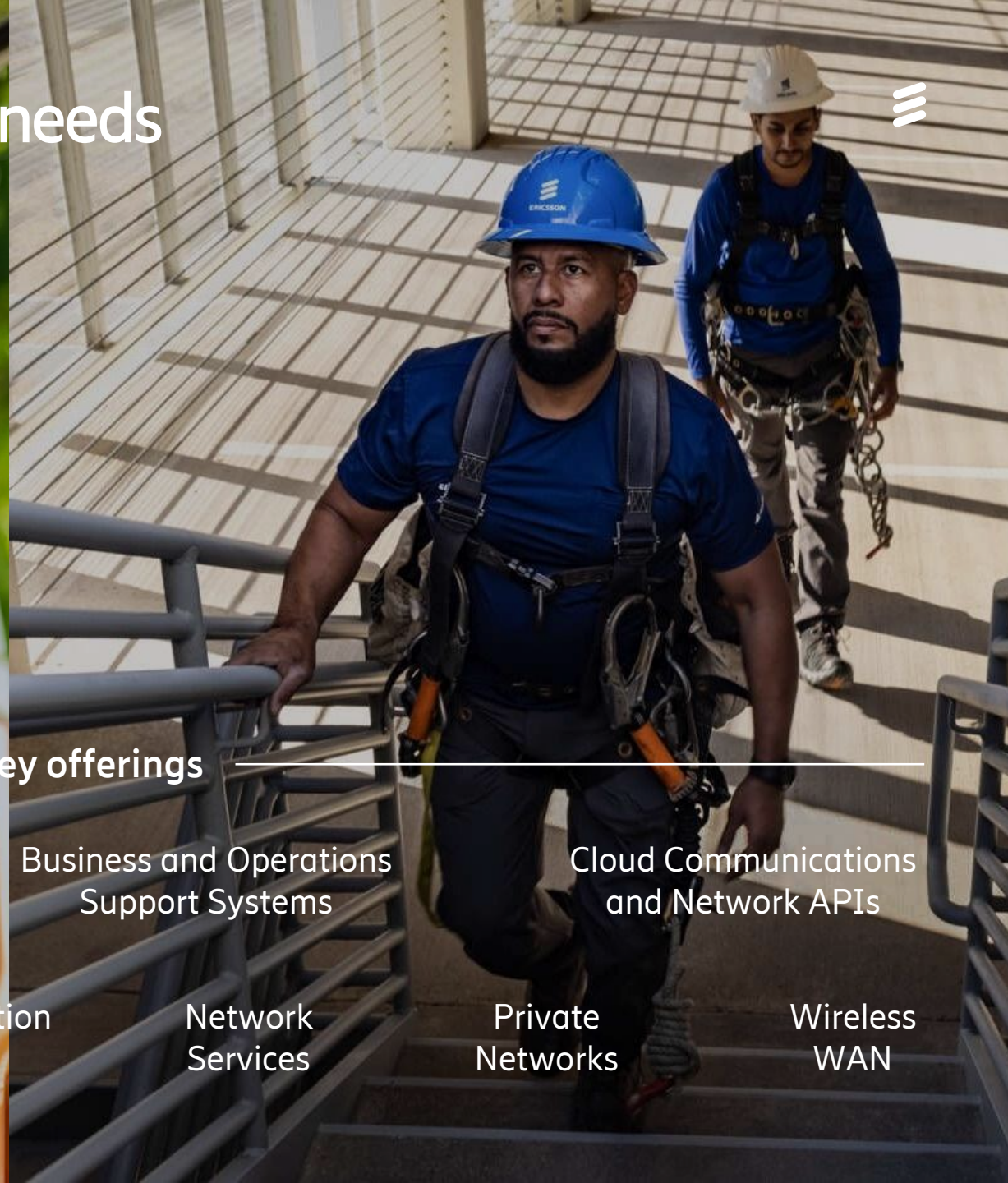


- Been at Ericsson for ~2 years
- Started my studies in 2020, graduated in 2025
- Took this course in 2024

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Offerings addressing our customers' needs

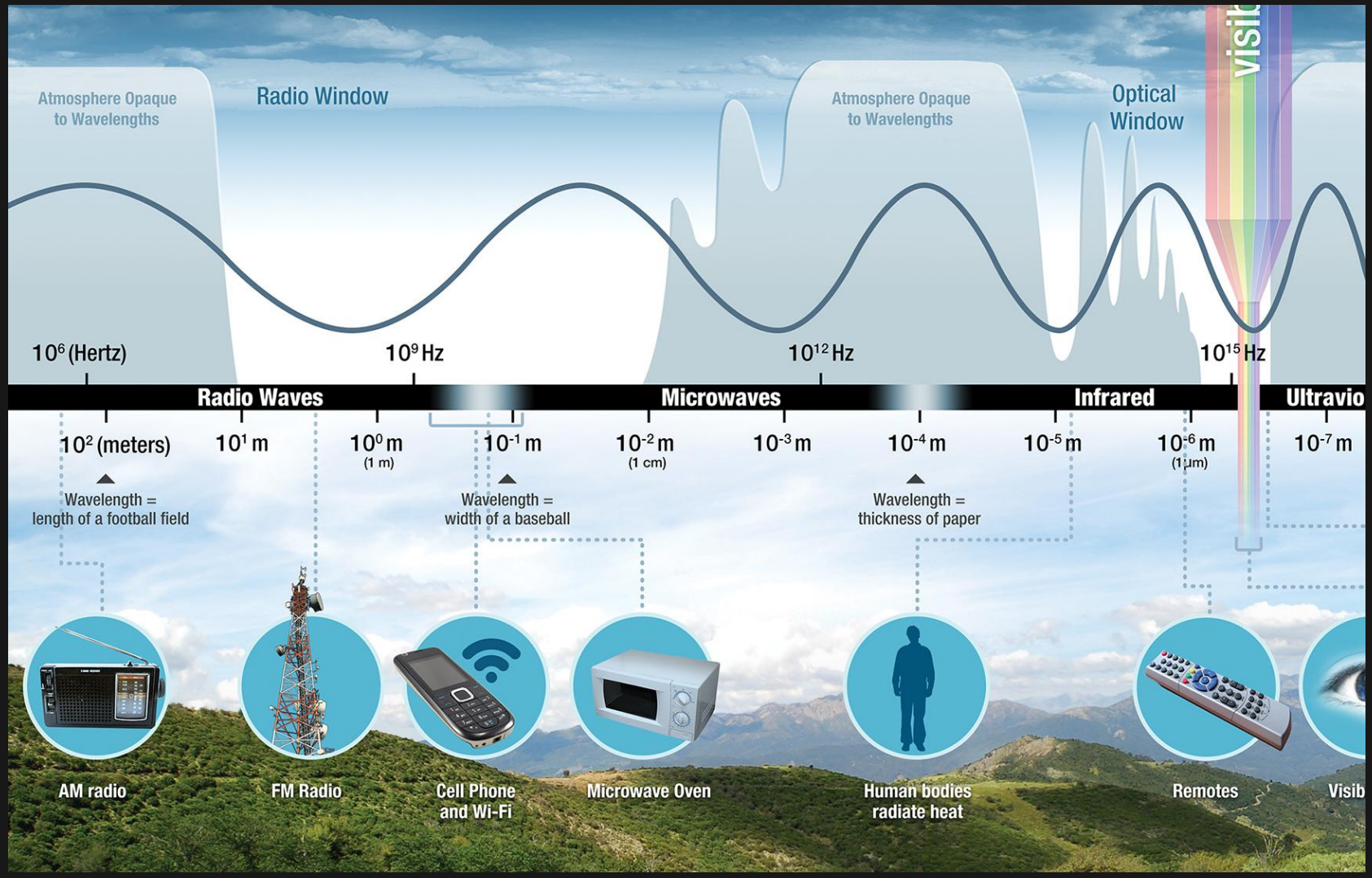




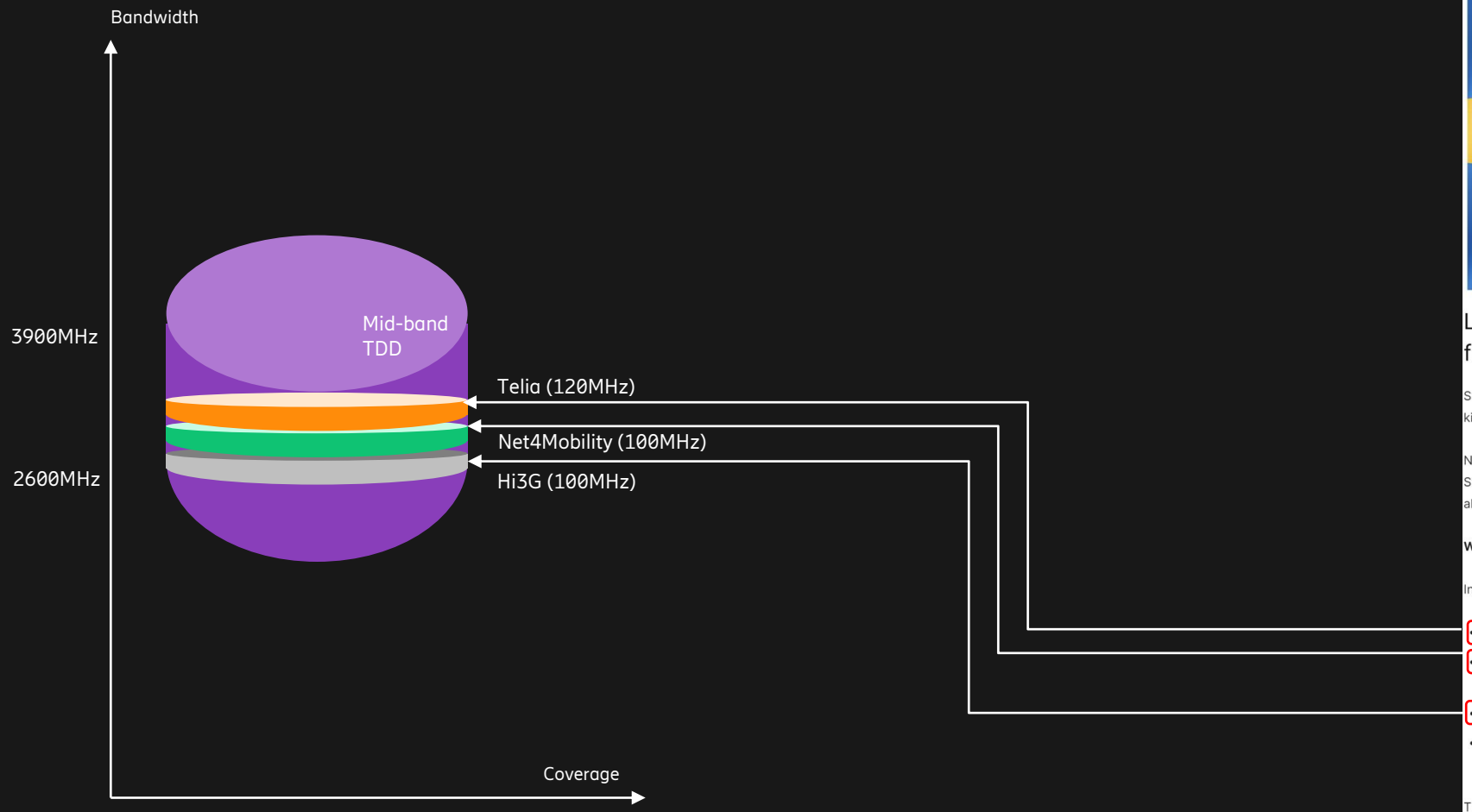
Ericsson key offerings

- 5G Core
- 5G Radio Access Network
- 5G Transport
- Business and Operations Support Systems
- Cloud Communications and Network APIs
- Managed Services
- Mission Critical Communications
- Network Automation and AI
- Network Services
- Private Networks
- Wireless WAN

Electromagnetic spectrum



Spectrum allocation



5G & Beyond | RAN | Access | News

Sweden completes spectrum auction in one day

By **Annie Turner** - 22 September 2023

Share



Licences were up for grabs in the 900MHz, 2.1GHz and 2.6GHz frequency ranges

Sweden's Post and Telecom Authority (PTS) announced the conclusion of its latest spectrum auction which kicked off on Wednesday.

Nordic telecom companies Tele2, Telenor Sweden and Telia Co have all acquired licences in the latest Swedish spectrum auction. They collectively invested SEK3.03 billion (€254.68 million) for spectrum allocations in the 900MHz, 2.1GHz and 2.6GHz auction.

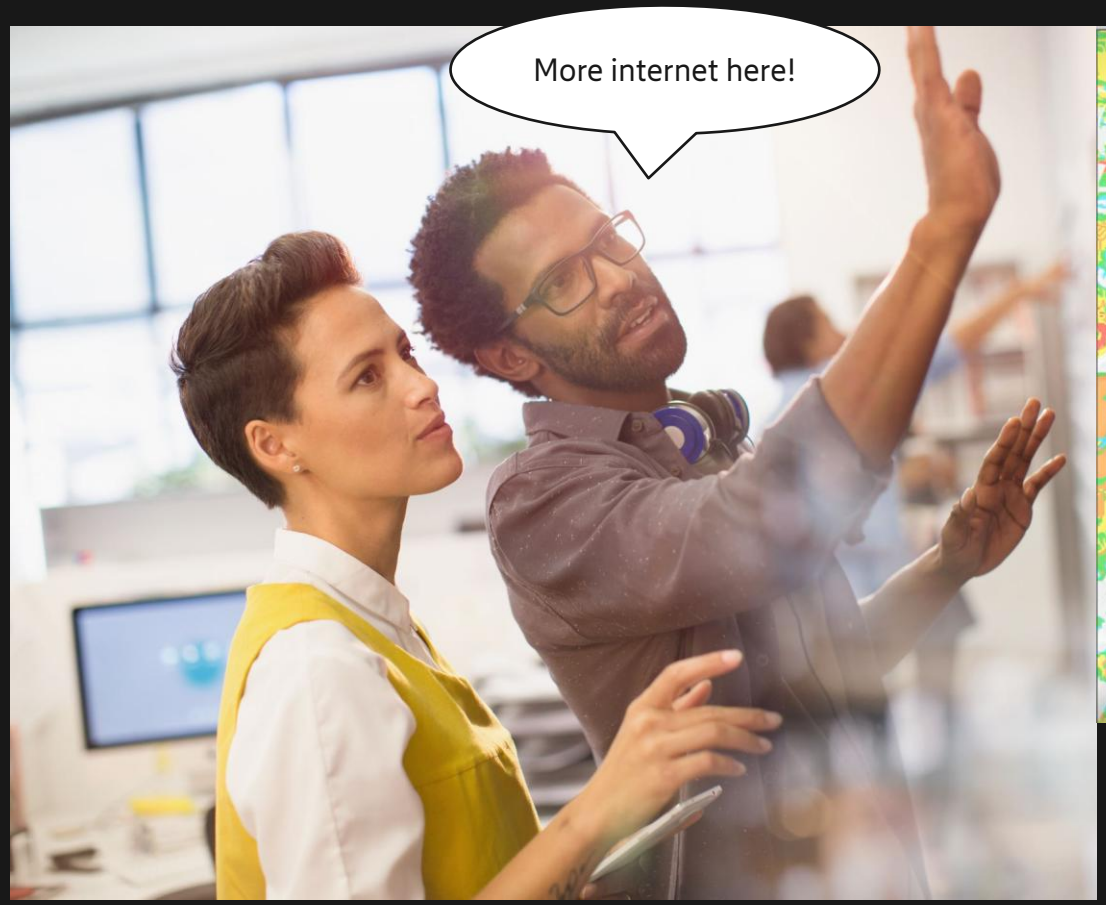
Who got what

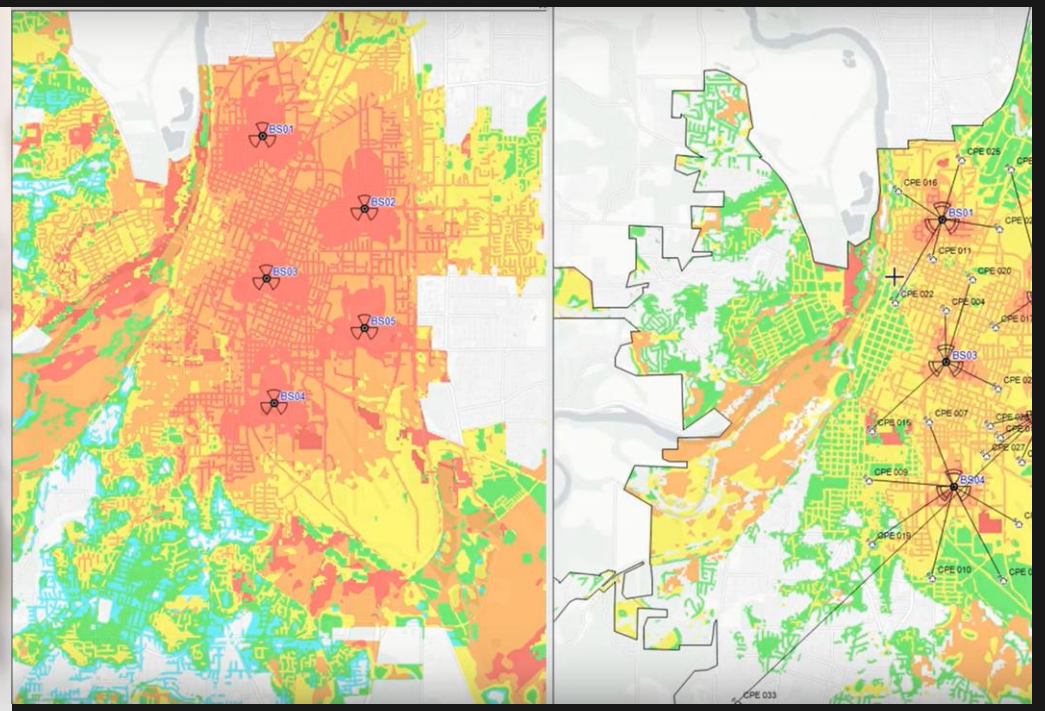
In a statement, PTS noted all 320 MHz at 3.5 GHz was assigned. Full allocations are as follows:

- Telia secured 120MHz (3500-3620 MHz) for SEK760.25 million SEK (€75 million)
- Net4Mobility (the joint venture between Tele2 and Telenor Sweden) won 100MHz (3620-3720MHz) for SEK665.5 million (€65 million)
- Hi3G secured 100MHz (3400-3500 MHz) at SEK491.25 million (€48 million)
- Teracom Group, which took over Net1 in 2019, won all the 80MHz on offer in the 2.3 GHz band for a total of SEK400 million (€40 million)

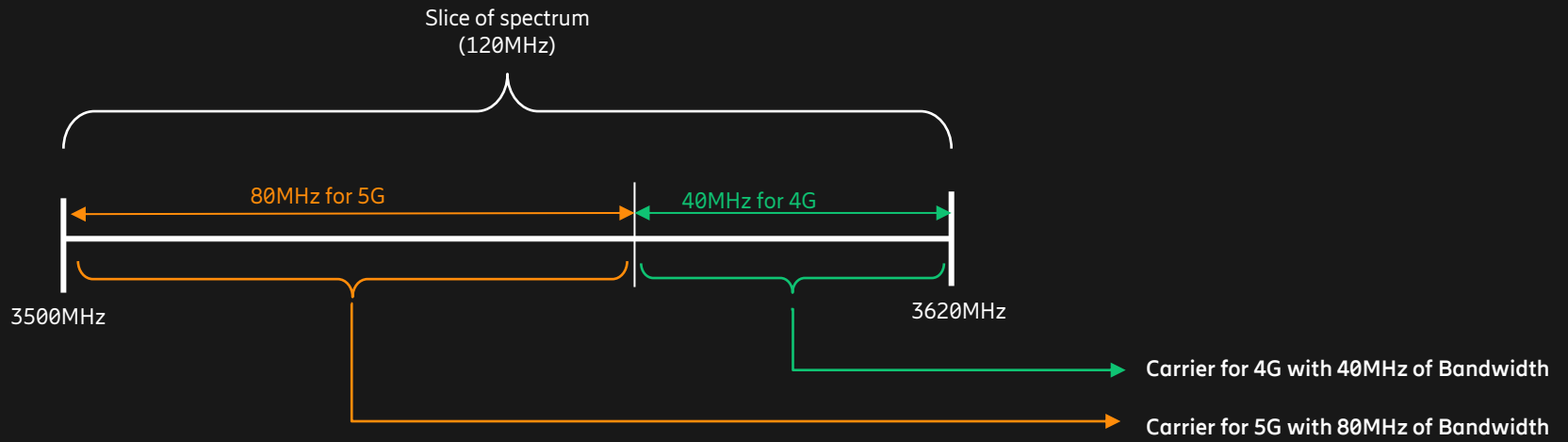
The four 3.5 GHz licences will be valid for a period of 25 years, from 20 January, 2021 to 31 December, 2045.

The network plan





Output of Planning Activity

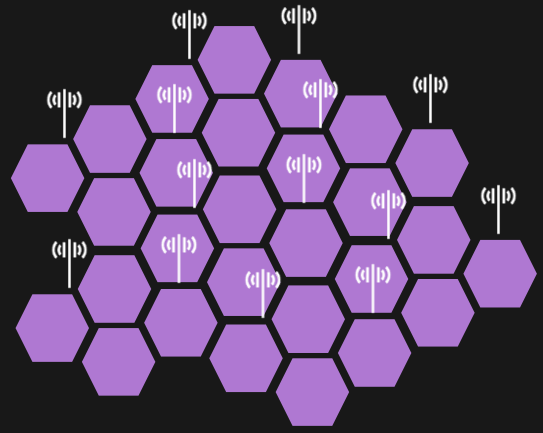


Option-1
- Full 120MHz for LTE (4G)

Option-2
- Full 120MHz for NR (5G)

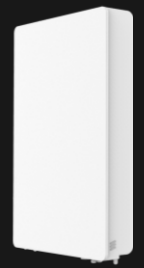
Option-3
- Mix of 4G & 5G
  - 60MHz for each
  - 40MHz for LTE & 80MHz for NR
  - And so on.....

Number of Sites





Radio ?? (3500-3700MHz)

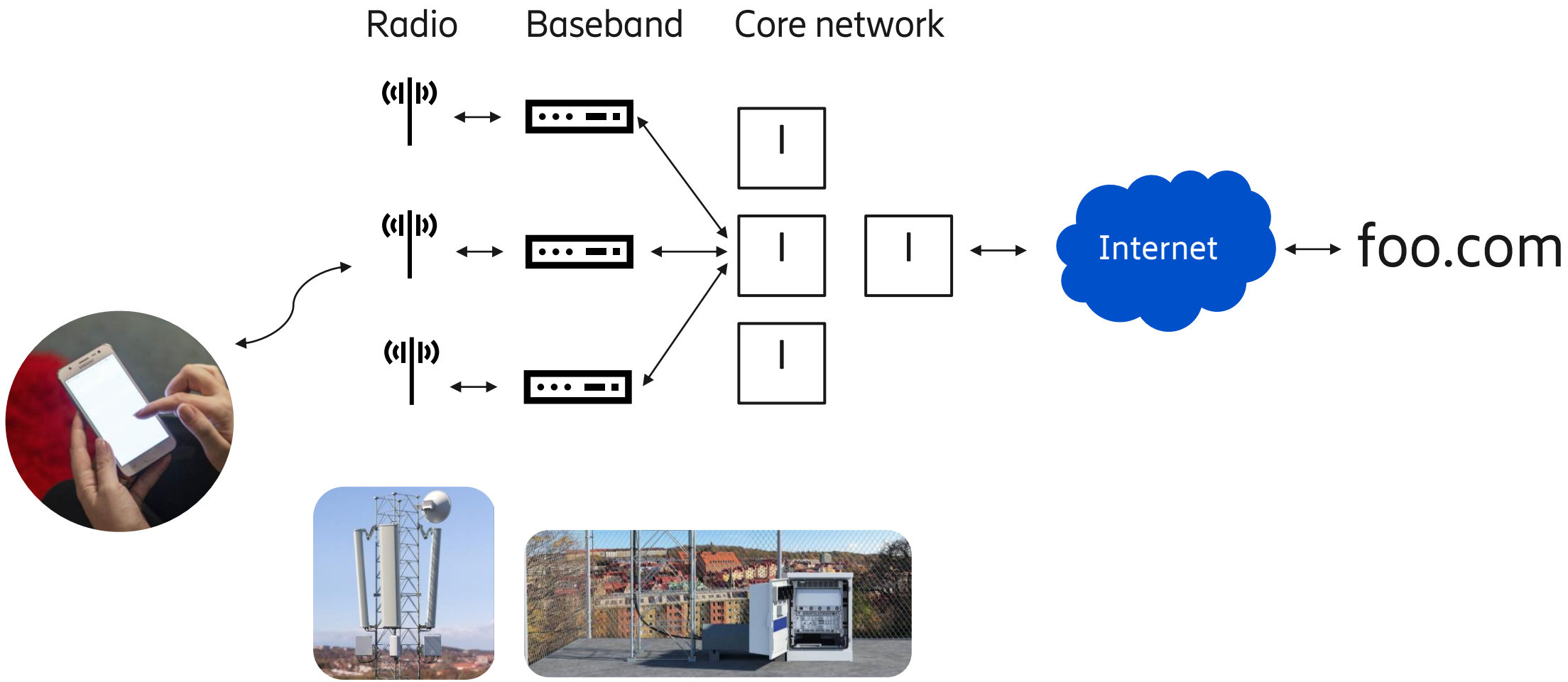


Radio ?? (3410-3800MHz)

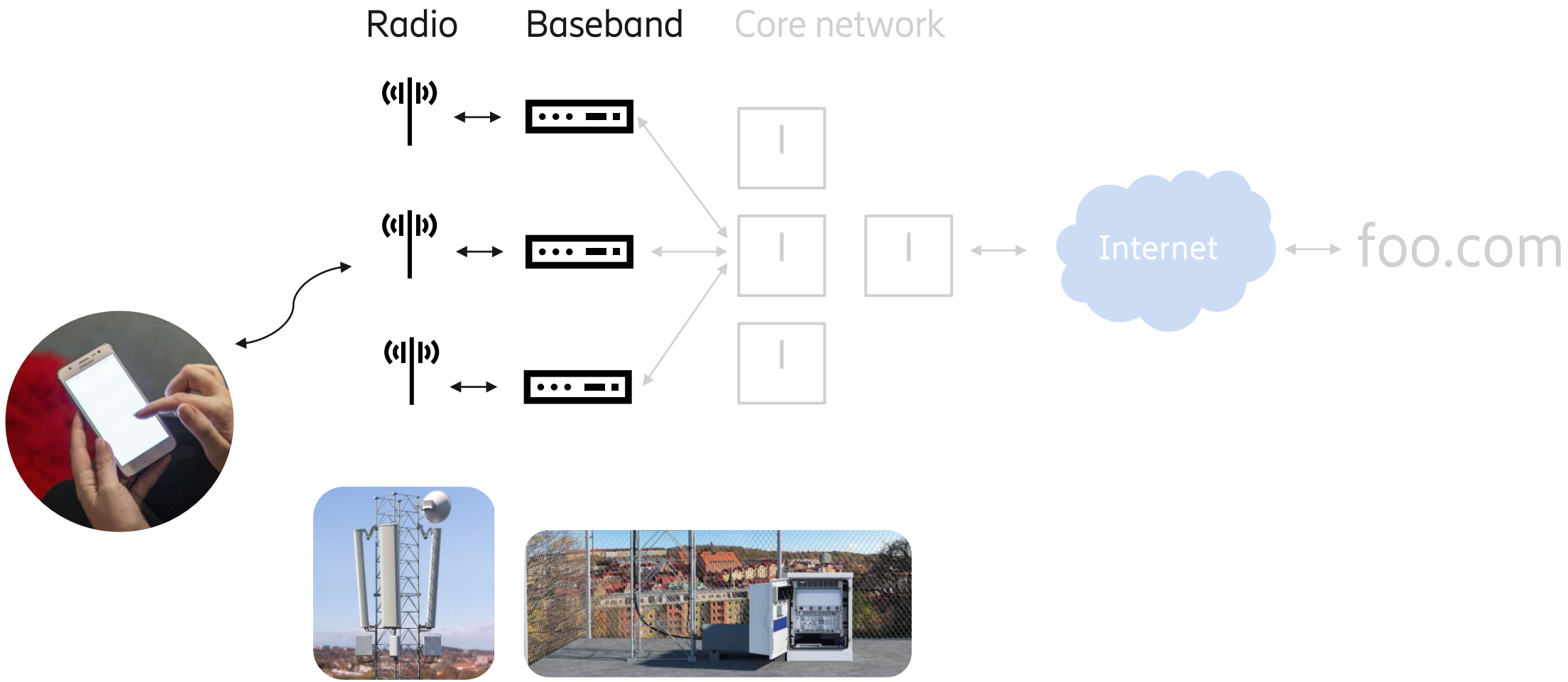


Radio ?? (3450-3800MHz)

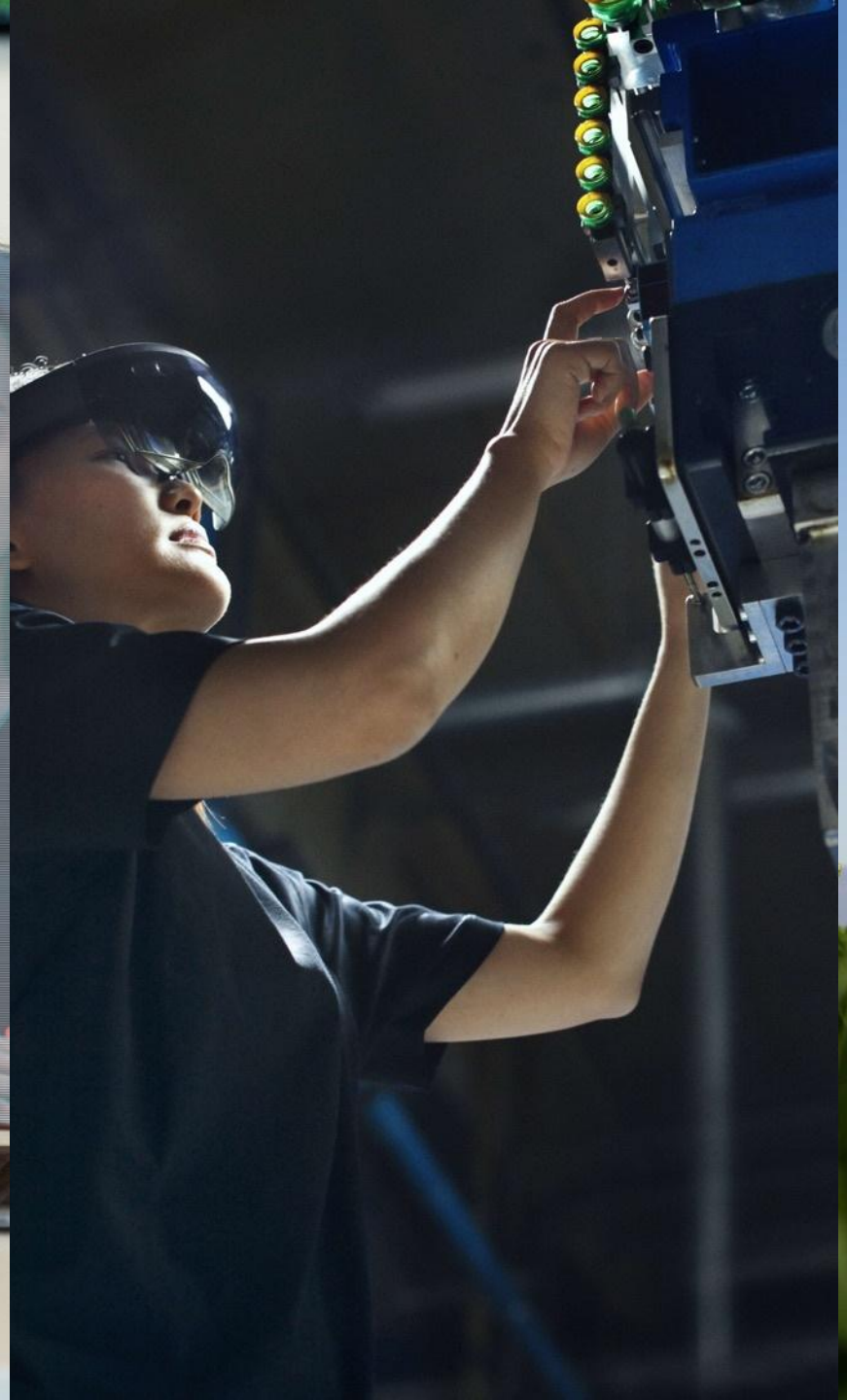
Radio Access Network (RAN) Overview



Radio Access Network (RAN) Overview







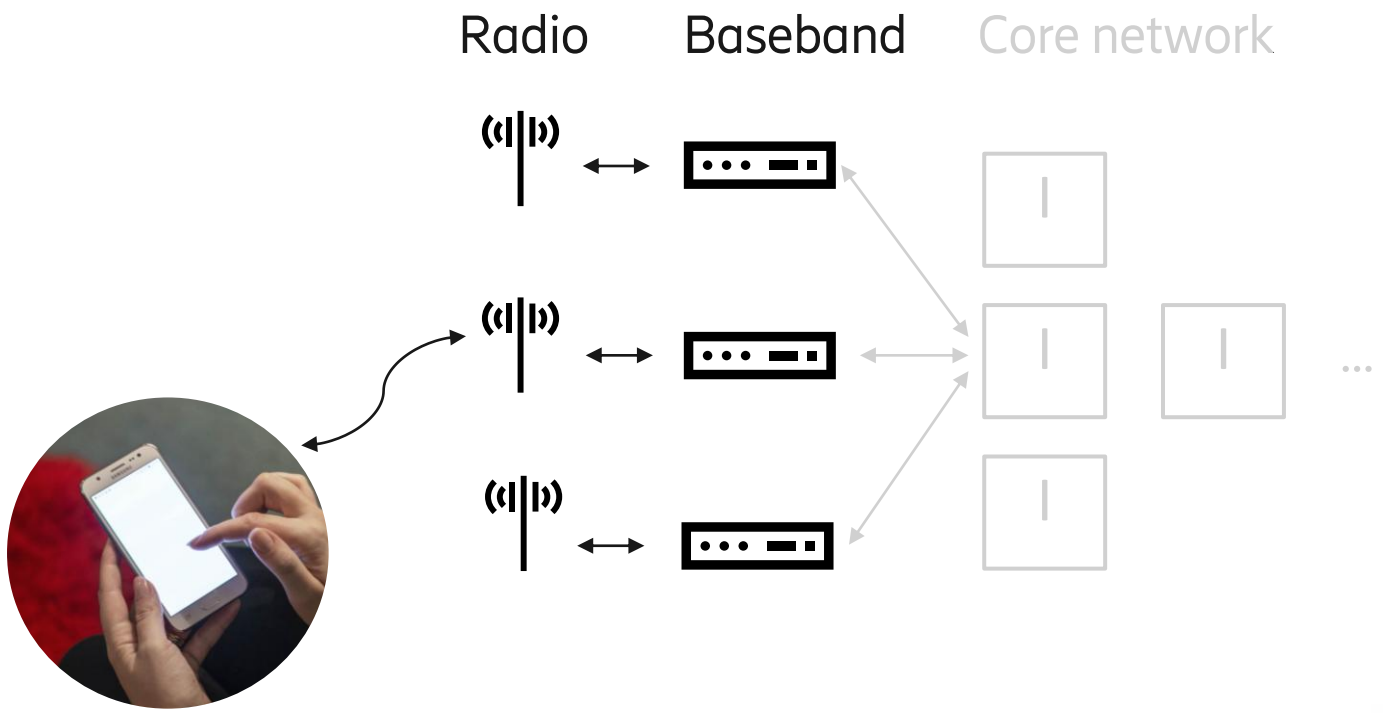


Product Configuration @ Ericsson

- Help sales with configuring products
- Configuration engines customize products to meet needs – like buying a couch
- When configured you get a list of all components that needs to be bought
  - Includes the non-obvious parts like screws etc



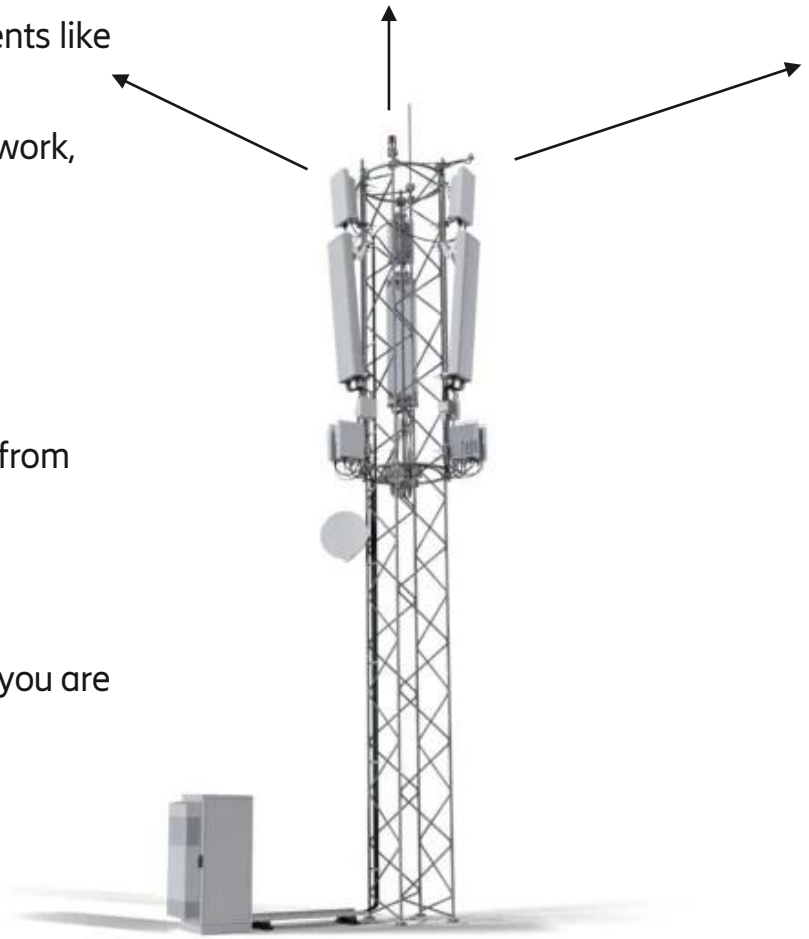
What are we configuring?





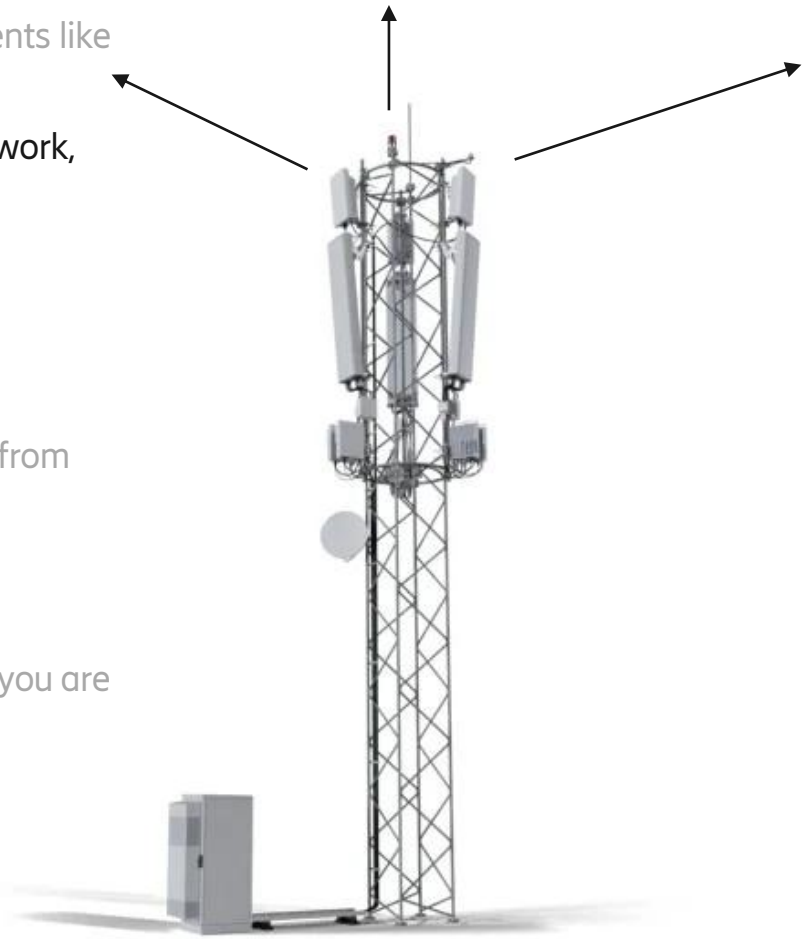
Product configuration – The site

- **Antenna System**: responsible for transmitting and receiving radio signals. It includes components like antennas, cables, and connectors.
- **Carrier:** range of frequencies allocated for transmitting and receiving signals on a wireless network, typically defined by its center frequency and bandwidth. (ephemeral)
- **RF Port:** interface that connects radio to antennas, split RX⬇/TX⬆ traffic.
- **Radio:** wireless communication component that transmits and receives radio signals.
- **CPRI Ports:** interface that connects radio to baseband, ⬇⬆ traffic.
- **Baseband:** network that handles the lower frequency signals, after they have been converted from radio frequencies (RF) by an antenna and receiver. It performs tasks such as switching, traffic management, timing, baseband processing, and radio interfacing.
- **Enclosure**: physical housing that protects the radio equipment
- **Fronthaul Components:** Routers used to connect the radio and baseband, mostly used when you are out of ports on the basebands or need a different port interface
- **Power Supply**: provides the electrical power to operate the radio solution
- **Transport Radio:** Sends wireless traffic between sites, instead of using fiber cables
- **Controller:** Used to wirelessly configure sites, without having to access them physically
- …



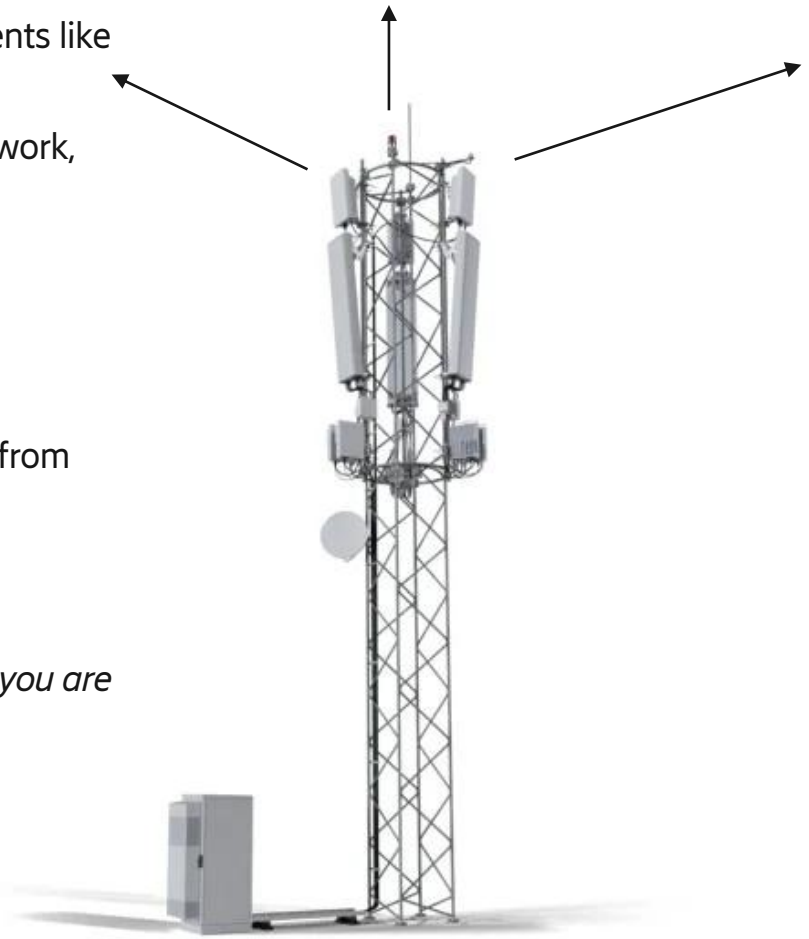
Product configuration – 2025

- **Antenna System**: responsible for transmitting and receiving radio signals. It includes components like antennas, cables, and connectors.
- **Carrier:** range of frequencies allocated for transmitting and receiving signals on a wireless network, typically defined by its center frequency and bandwidth. (ephemeral)
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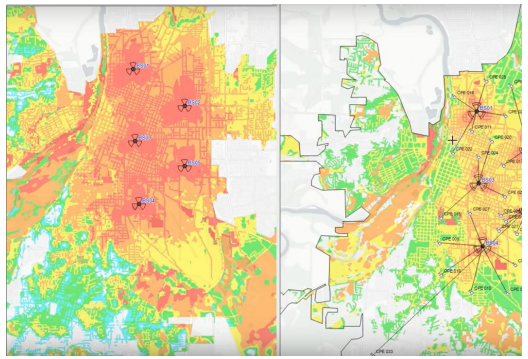


Product configuration – Today

- **Antenna System**: responsible for transmitting and receiving radio signals. It includes components like antennas, cables, and connectors.
- **Carrier:** range of frequencies allocated for transmitting and receiving signals on a wireless network, typically defined by its center frequency and bandwidth. (ephemeral)
- **RF Port:** interface that connects radio to antennas, split RX⬇/TX⬆ traffic.
- **Radio:** wireless communication component that transmits and receives radio signals.
- **CPRI Ports:** interface that connects radio to baseband, ⬇⬆ traffic.
- **Baseband:** network that handles the lower frequency signals, after they have been converted from radio frequencies (RF) by an antenna and receiver. It performs tasks such as switching, traffic management, timing, baseband processing, and radio interfacing.
- **Enclosure**: physical housing that protects the radio equipment
- ***Fronthaul Components:*** *Routers used to connect the radio and baseband, mostly used when you are out of ports on the basebands or need a different port interface*
- **Power Supply**: provides the electrical power to operate the radio solution
- **Transport Radio:** Sends wireless traffic between sites, instead of using fiber cables
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- …

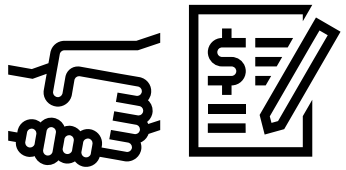


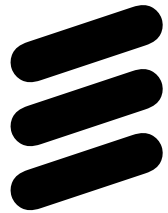
The scenario – Network planning

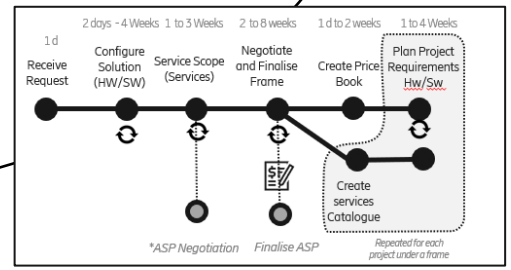


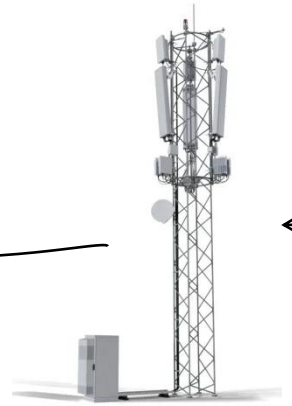


The customer



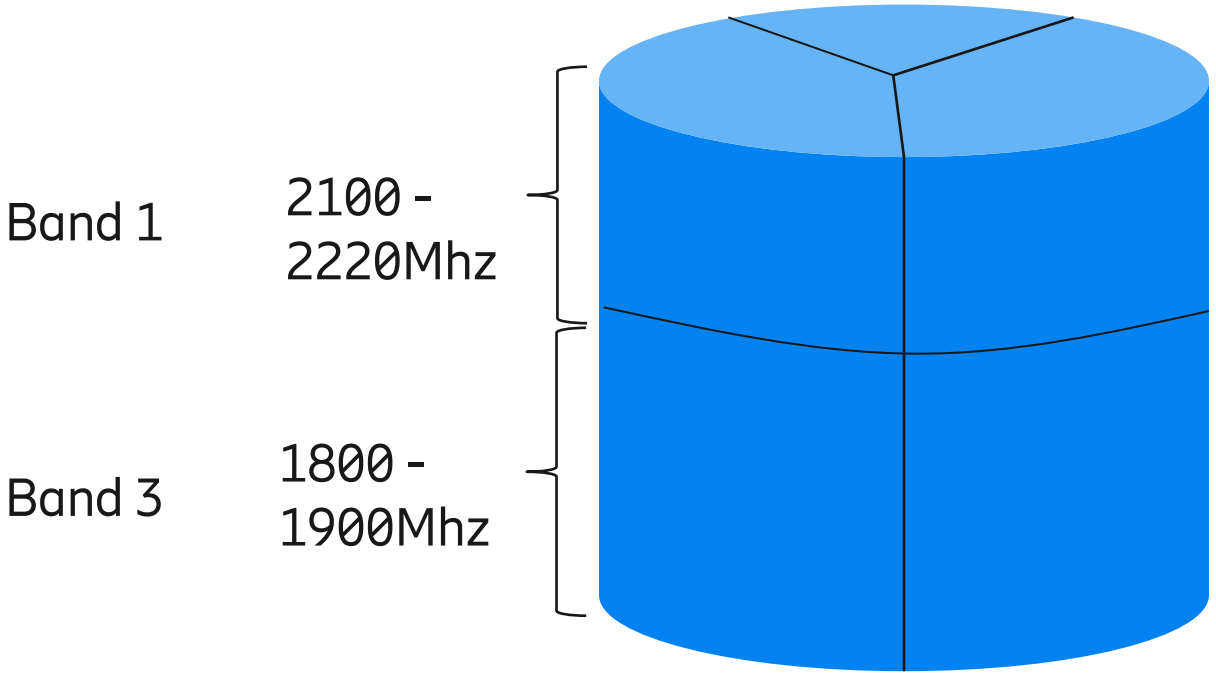




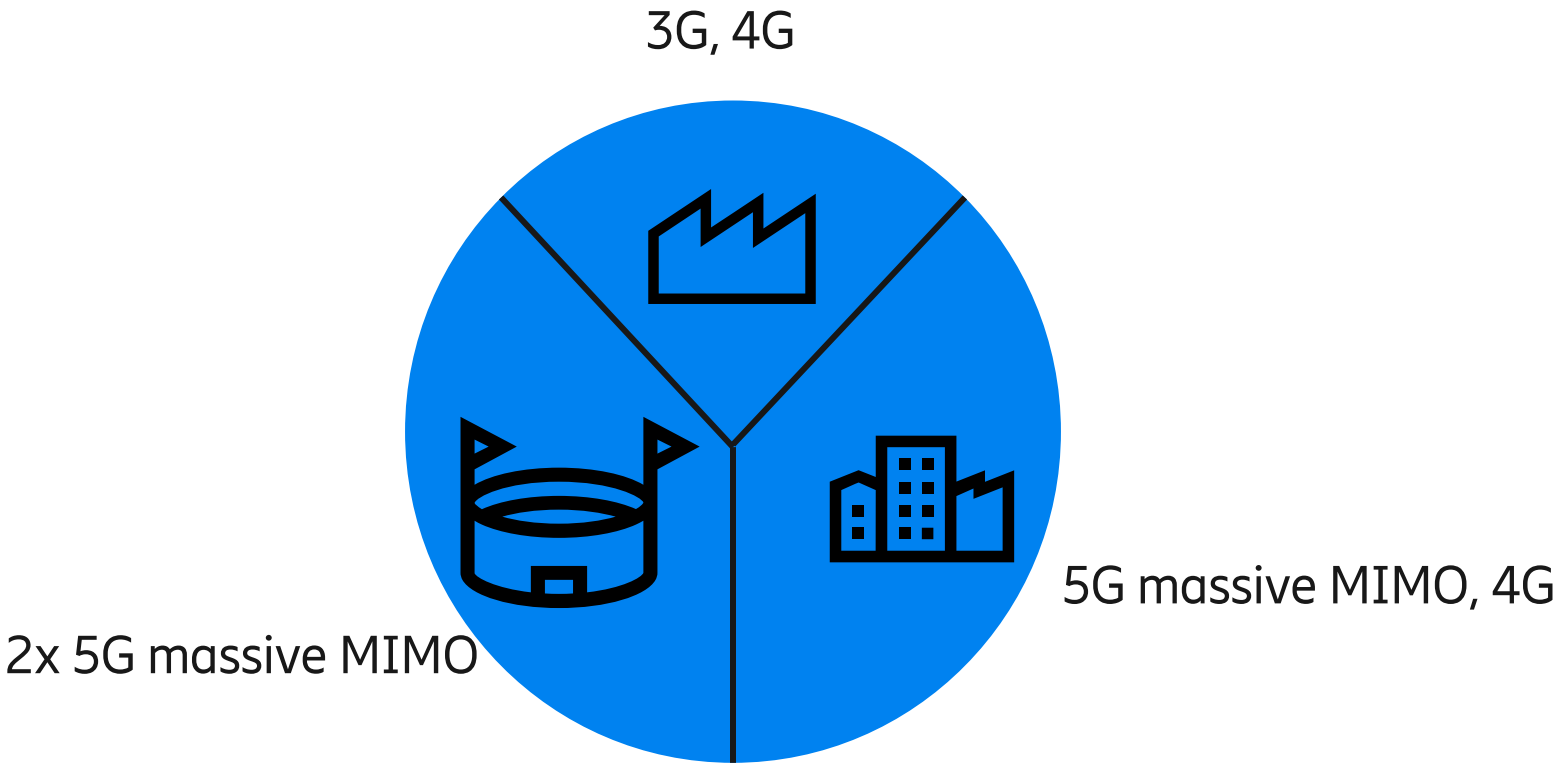


*The scenario is extremely simplified, the process is more complex than described

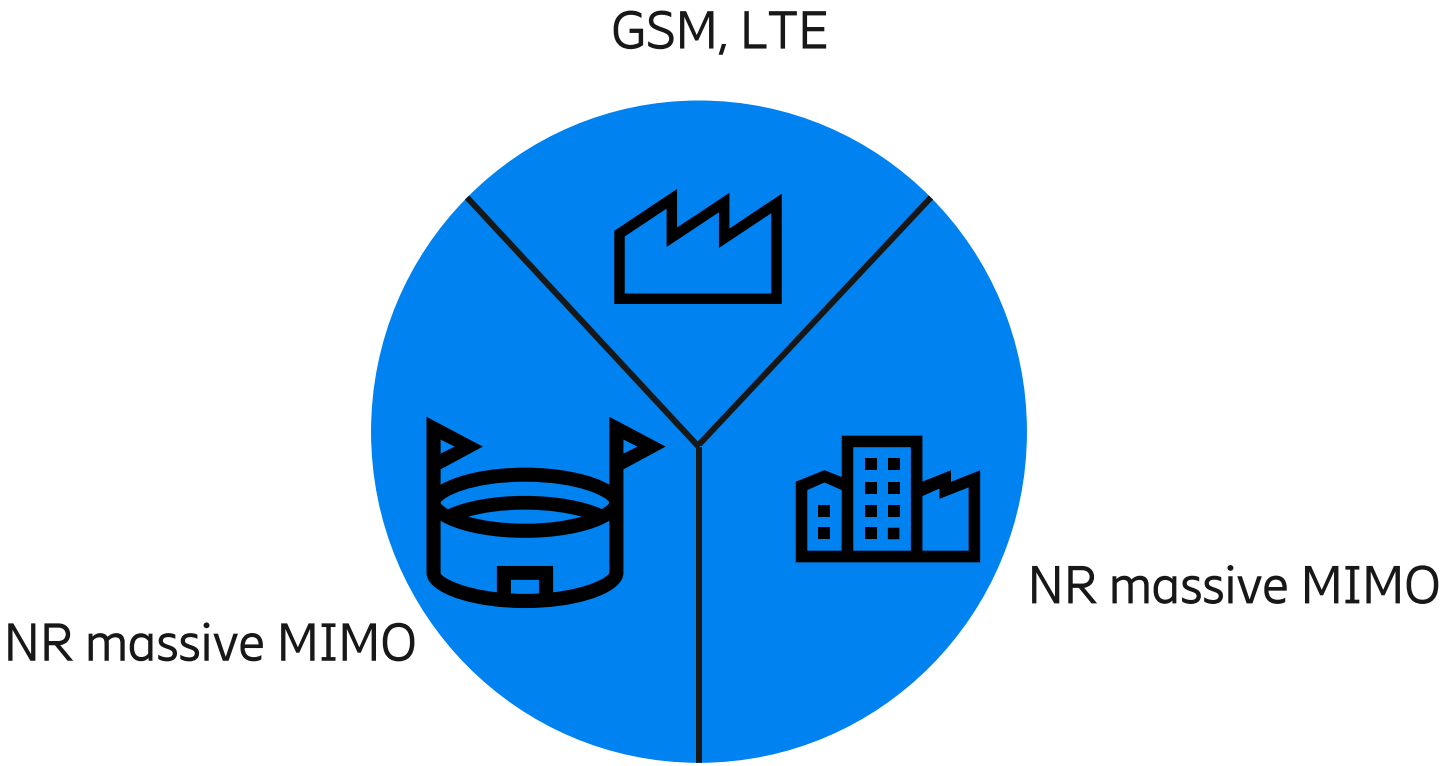
Network planning – The requirements



Radio Solution– The requirements



Radio Solution– The requirements



Product configuration – Baseband Dimensioning

Resource allocation - a combinatorial optimization problem

- Baseband Dimensioning:
  - Carriers ⇆ Baseband
- Solution:
  - ~40 table constraints (compatibilities)
  - Bin-packing (load channeling)
  - Linear constraints (capacities)

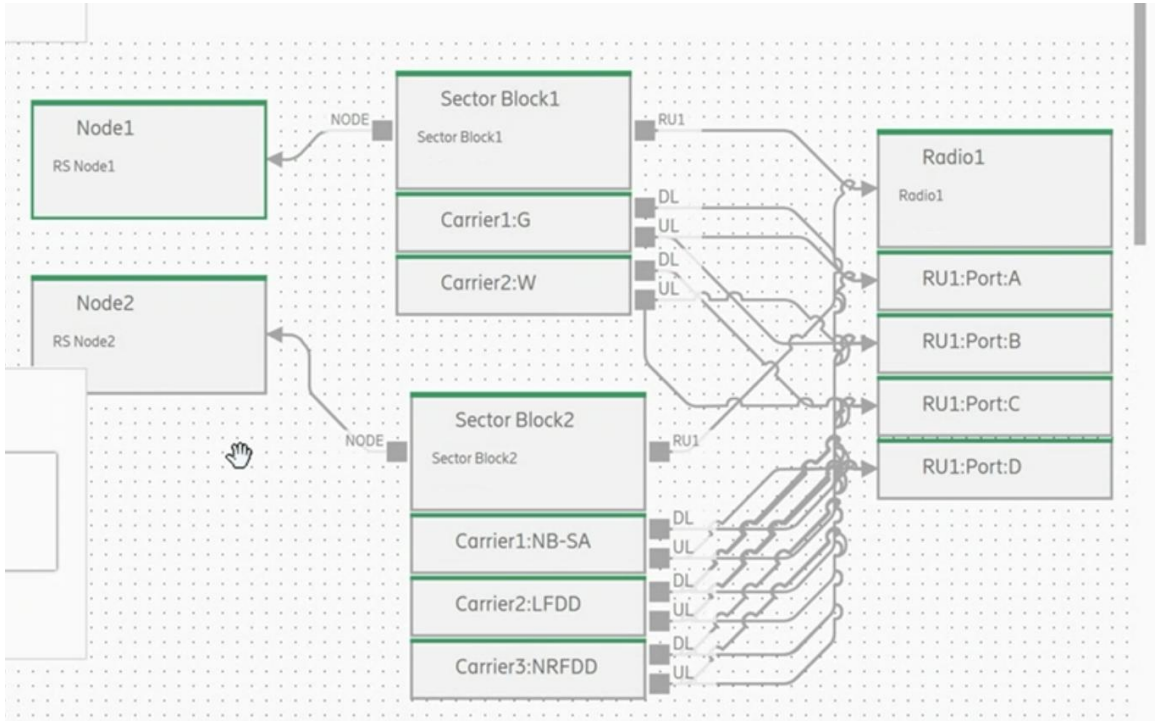


Given a set of carriers C, select a set of basebands B and map every carrier $c \in C$ to a baseband $b \in B$ such that all compatibility and capacity constraints of the specified basebands are met.

Product configuration – Radio solution

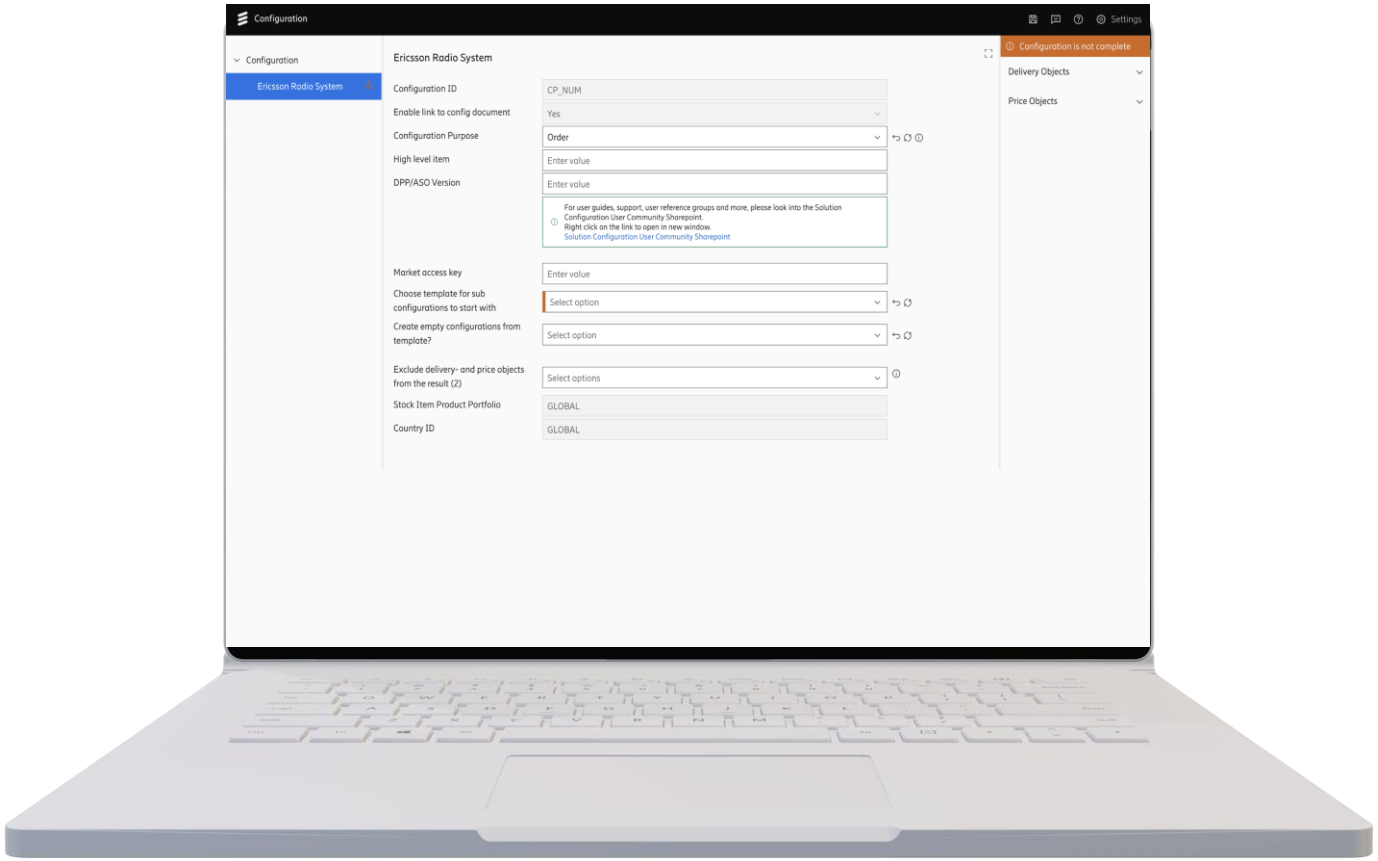
Resource allocation - a combinatorial optimization problem

- Radio Solution:
  - Carriers ⇆ Radios
- Solution:
  - ~60 table constraints (compatibilities)
  - Bin-packing (e.g. connecting cables)
  - Linear constraints (capacities)
  - Value-precede chain
  - …



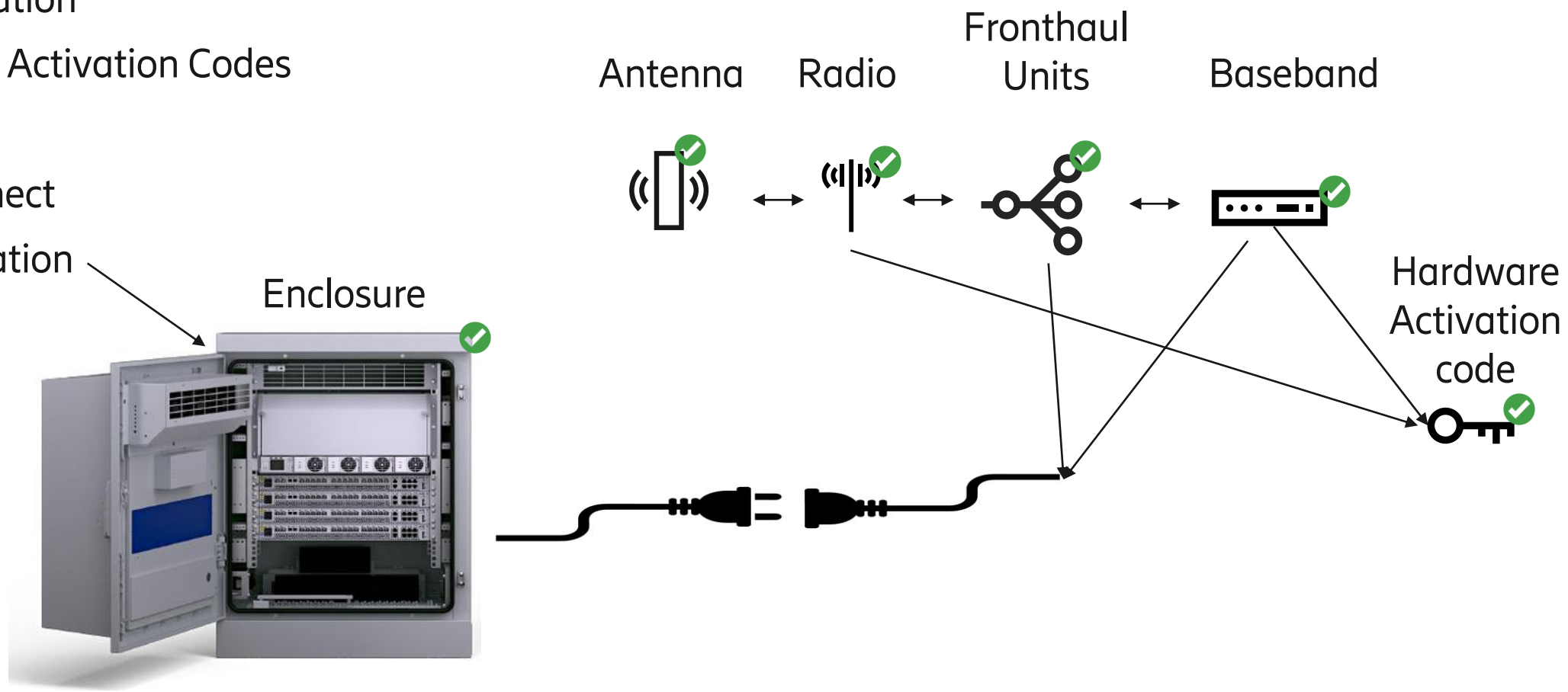
Given a set C of carriers, select a set R of radio units and map every carrier $c \in C$ to a radio $r \in R$ such that r meets all demands of c mapped to r and the capacities of r are not exceeded.

Demo


Configuration
Settings
Configuration
Ericsson Radio System
Ericsson Radio System
Configuration ID
CP_NUM
Enable link to config document
Yes
Configuration Purpose
Order
High level item
Enter value
DPP/ASO Version
Enter value
For user guides, support, user reference groups and more, please look into the Solution Configuration User Community Sharepoint.
Right click on the link to open in new window.
Solution Configuration User Community Sharepoint
Market access key
Enter value
Choose template for sub configurations to start with
Select option
Create empty configurations from template?
Select option
Exclude delivery- and price objects from the result (2)
Select options
Stock Item Product Portfolio
GLOBAL
Country ID
GLOBAL
Configuration is not complete
Delivery Objects
Price Objects

Product configuration – Model Portfolio

- Baseband Dimensioning
- Radio Solution
- Hardware Activation Codes
- Antenna
- RAN Connect
- Rail Allocation



Product configuration – Optimization

- Default – make the configuration as *tight* as possible
- Lexicographic
  - First optimize on f_1 then on f_2 etc
  - Keep f_1 as low as possible, f_2 should also be minimized but f_1 should still be at its lowest point
- RadioSolution - Multi-objective, minimize:
  - Number of Radios
  - Number of Ports
  - …

Radio Solution – A Large MiniZinc Model



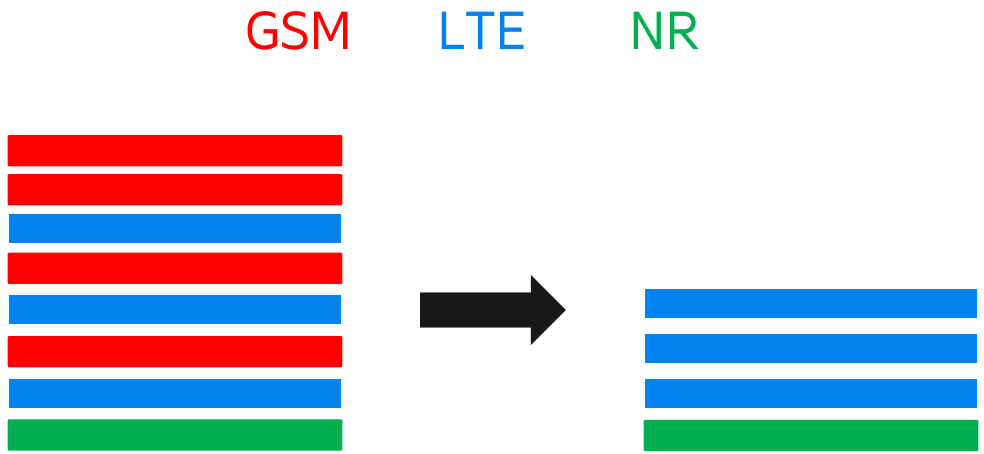
The Radio Solution Model

- 149 Constraints
- 143 Arrays of Decision Variables
- ~11k LoC
- 97 Data tables
- Top 3 largest tables #rows:
  - 22,661
  - 20,309
  - 17,701
- Solving
  - **CP-SAT 8 cores**
  - Huub
  - Gecode

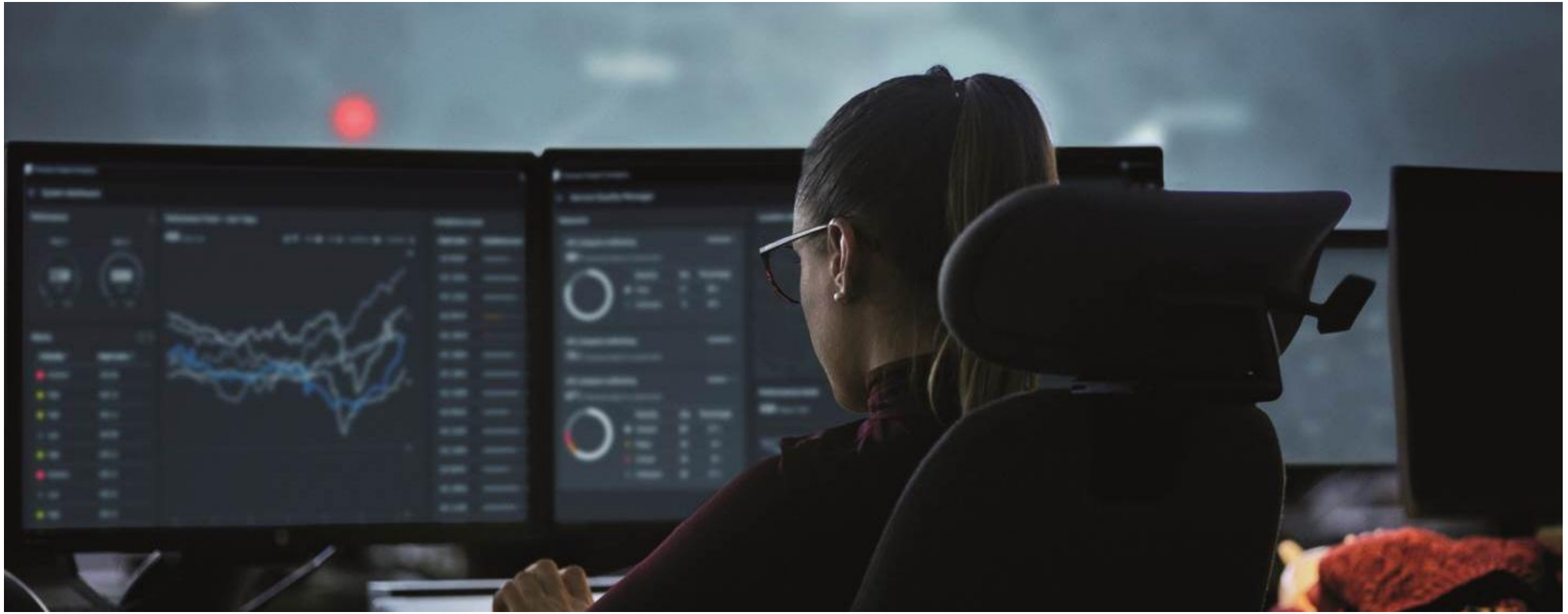


Pre-filtering

- Lots of data, lots of radios to choose from
  - Based on the instance, filter certain radios
- Create sub tables
  - "If there is no GSM carrier for this instance, why have data for it in the tables"
- Drastically decreases solve time

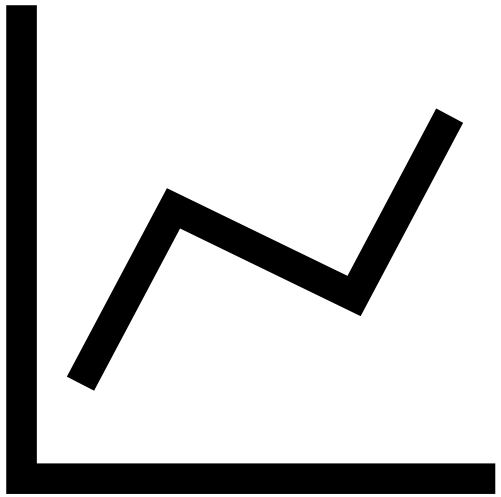


Maintaining our MiniZinc model



Make It Work
Make It Right
Make It Fast

Ever increasing complexity over the last 3 years

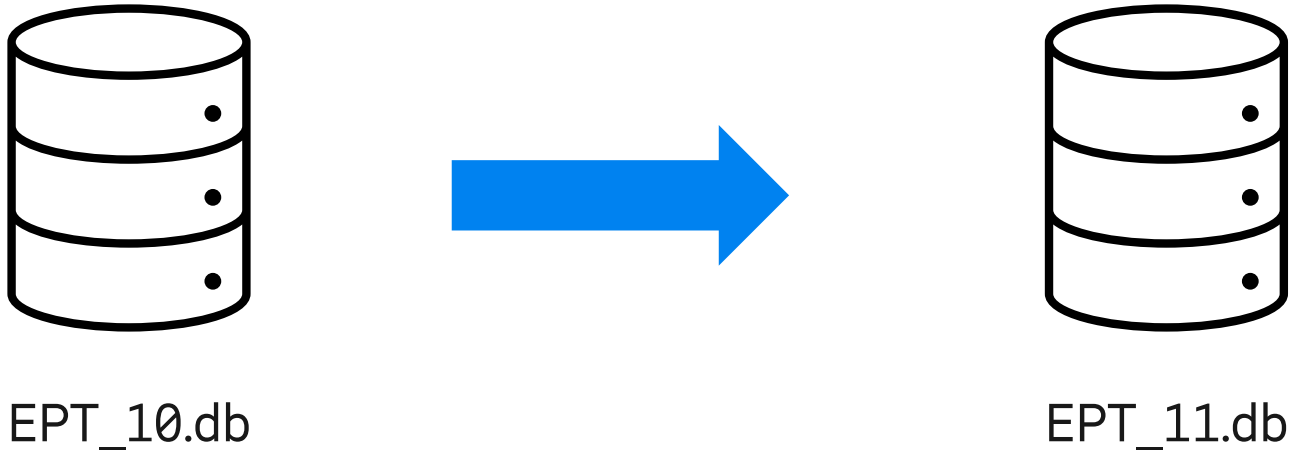


- +4K LoC *.py → +8K LoC *.py
- +4K LoC *.mzn → +11K LoC * .mzn
- +7K LoC *.rst → +11K LoC *.rst
- +60K LoC *.dzn → +177K LoC *.dzn
- 30 testcases → ~140 testcases

- New products
- New rules
- New language features

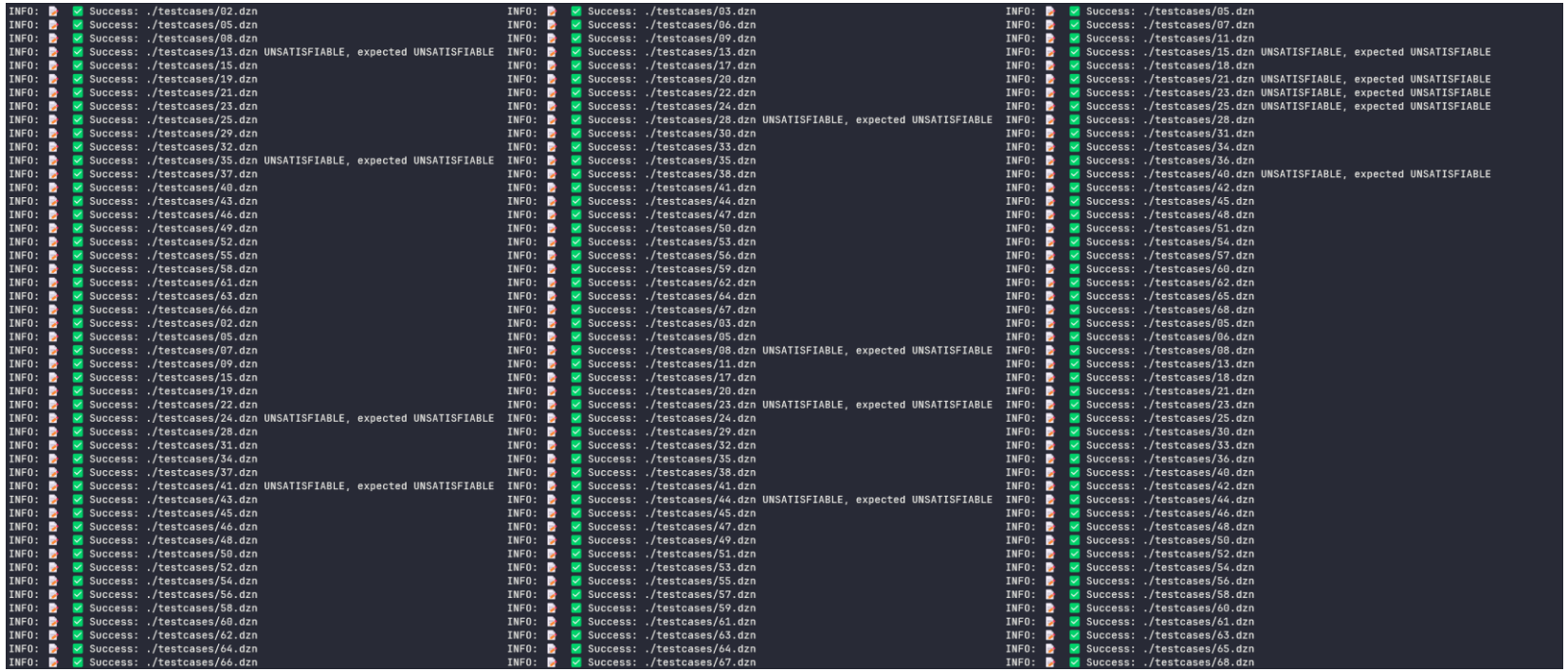
Living Data / Monthly Release

- New Radios/Baseband/Routers
  - New Hardware revision
  - New Software revision
- New or Reworks of Rules
  - New table
- EOS for Units



More test + integration

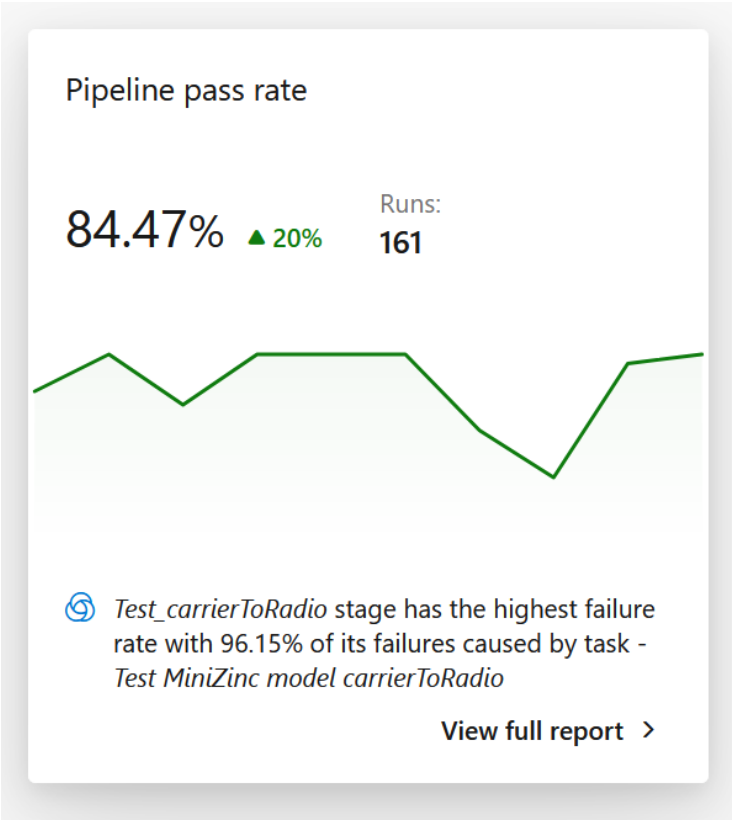
Better test coverage, less regressions & bugs



- Testing both SAT/UNSAT
- New rules
  -> new test
- New bug/regression
  -> new test

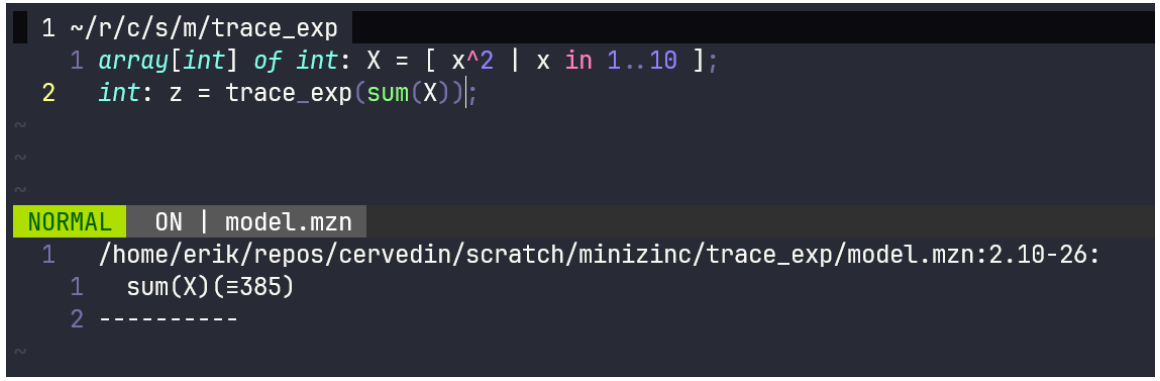
Challenges – Testing

- Hard to test specific parts of the model
  - Need to test the whole model
  - Test using partial assignments
  - Test both for SAT/UNSAT
  - Indirect tests
- Rigorous testing
  - Automated testing (CI/CD/MLOPS)
  - Improving coverage (removing constraint should fail)
  - Large changes difficult (testcase might be symmetrical)



Challenges – Debugging

- Conditional debug output
- `trace_exp`
  - Prints an expression and value
- Black box
  - Poke the box and see what happens!
    - Manual assignments
    - Manually bisect constraints (delete/comment) until UNSAT becomes SAT



What's next?

- Better test framework
- Object-oriented MiniZinc?
- Project based configuration
  - Configuring 100 sites at the same time
- New models
  - Fronthaul dimensioning


Imagine Possible
ericsson.com/careers

Questions?